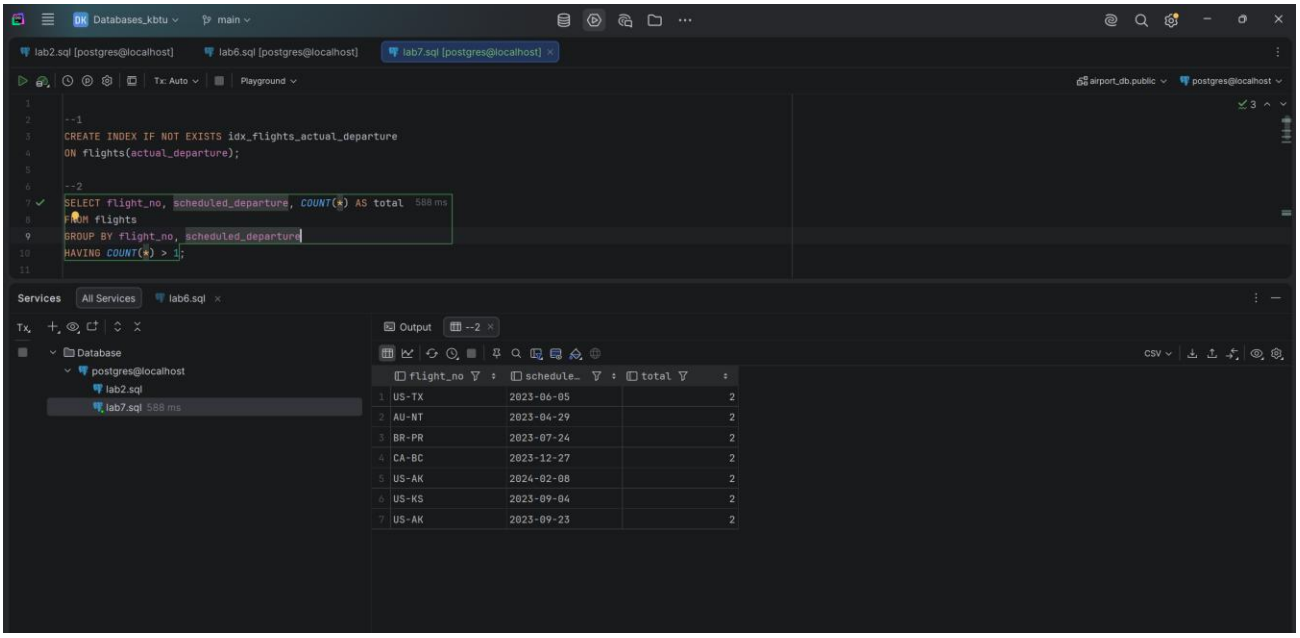
```
--1
CREATE INDEX IF NOT EXISTS idx_flights_actual_departure  113 ms
ON flights(actual_departure);
```

2--

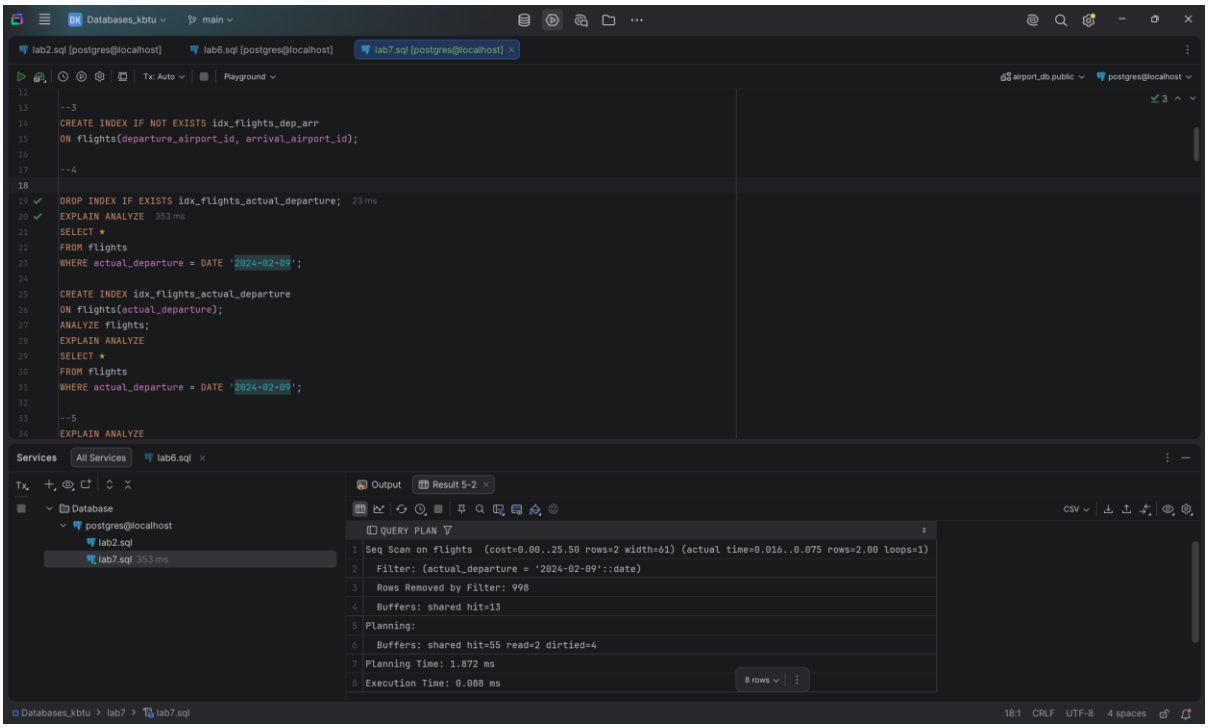


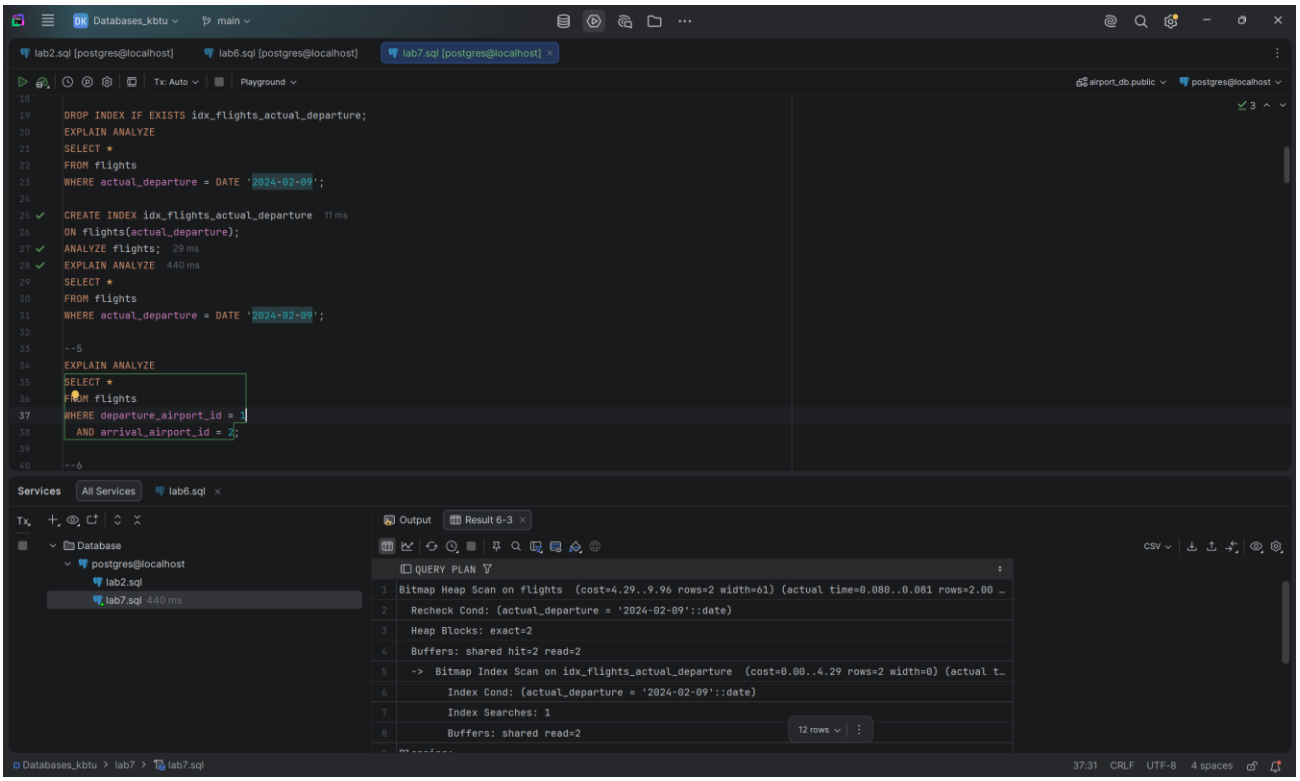
3—

```
--3
CREATE INDEX IF NOT EXISTS idx_flights_dep_arr  17 ms
ON flights(departure_airport_id, arrival_airport_id);
```

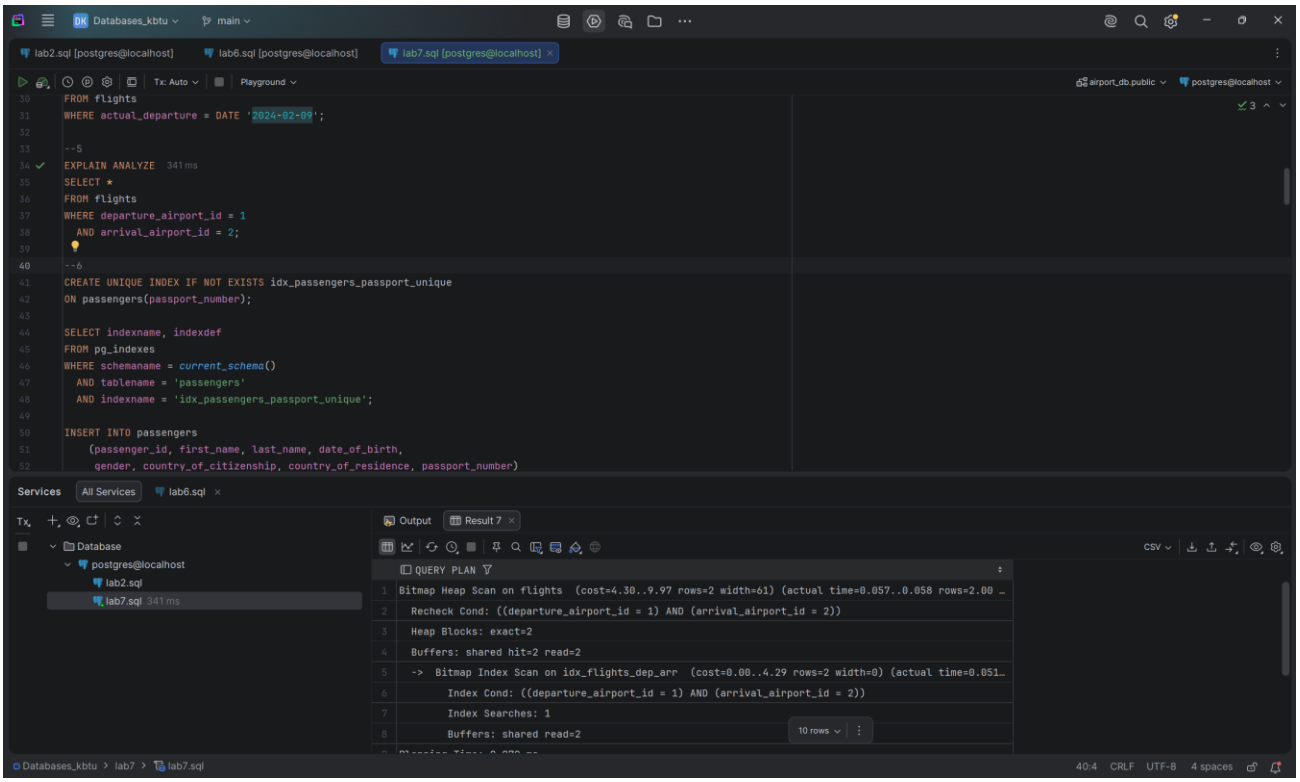
\

4—

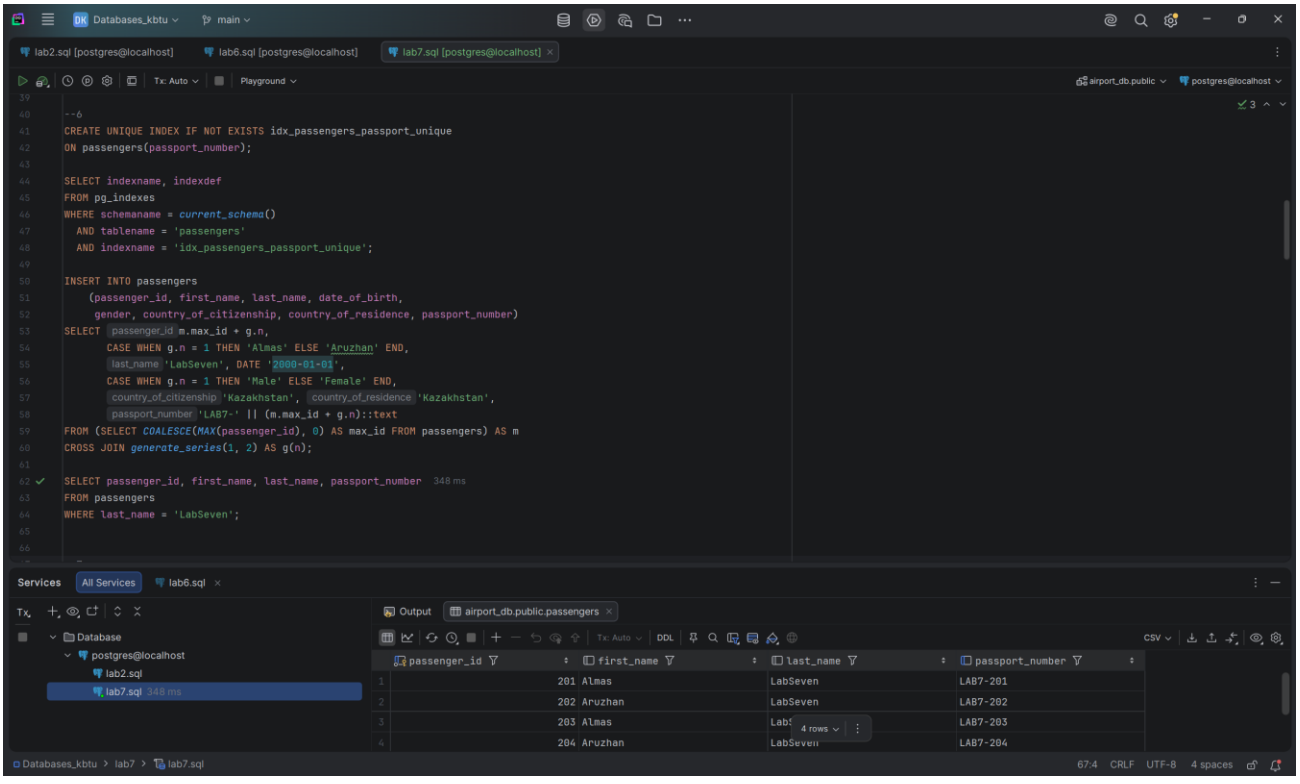

Databases_kbtu
main
lab2.sql [postgres@localhost]
lab6.sql [postgres@localhost]
lab7.sql [postgres@localhost]
Tx: Auto
Playground
airport_db.public
postgres@localhost
--3
CREATE INDEX IF NOT EXISTS idx_flights_dep_arr
ON flights(departure_airport_id, arrival_airport_id);
--4
DROP INDEX IF EXISTS idx_flights_actual_departure; 23 ms
EXPLAIN ANALYZE 353 ms
SELECT *
FROM flights
WHERE actual_departure = DATE '2024-02-09';
CREATE INDEX idx_flights_actual_departure
ON flights(actual_departure);
ANALYZE flights;
EXPLAIN ANALYZE
SELECT *
FROM flights
WHERE actual_departure = DATE '2024-02-09';
--5
EXPLAIN ANALYZE
Services
All Services
lab6.sql
Database
postgres@localhost
lab2.sql
lab7.sql 353 ms
Output
Result 5-2
CSV
QUERY PLAN
Seq Scan on flights (cost=0.00..25.50 rows=2 width=61) (actual time=0.016..0.075 rows=2.00 loops=1)
Filter: (actual_departure = '2024-02-09'::date)
Rows Removed by Filter: 998
Buffers: shared hit=13
Planning:
Buffers: shared hit=55 read=2 dirtied=4
Planning Time: 1.872 ms
Execution Time: 0.088 ms
8 rows
Databases_kbtu > lab7 > lab7.sql
18:1 CRLF UTF-8 4 spaces


Databases_kbtu
main
lab2.sql [postgres@localhost]
lab6.sql [postgres@localhost]
lab7.sql [postgres@localhost]
Tx: Auto
Playground
airport_db.public
postgres@localhost
DROP INDEX IF EXISTS idx_flights_actual_departure;
EXPLAIN ANALYZE
SELECT *
FROM flights
WHERE actual_departure = DATE '2024-02-09';
CREATE INDEX idx_flights_actual_departure 11 ms
ON flights(actual_departure);
ANALYZE flights; 29 ms
EXPLAIN ANALYZE 440 ms
SELECT *
FROM flights
WHERE actual_departure = DATE '2024-02-09';
--5
EXPLAIN ANALYZE
SELECT *
FROM flights
WHERE departure_airport_id = 1
AND arrival_airport_id = 2;
--6
Services
All Services
lab6.sql
Database
postgres@localhost
lab2.sql
lab7.sql 440 ms
Output
Result 6-3
CSV
QUERY PLAN
Bitmap Heap Scan on flights (cost=4.29..9.96 rows=2 width=61) (actual time=0.080..0.081 rows=2.00 …
Recheck Cond: (actual_departure = '2024-02-09'::date)
Heap Blocks: exact=2
Buffers: shared hit=2 read=2
-> Bitmap Index Scan on idx_flights_actual_departure (cost=0.00..4.29 rows=2 width=0) (actual t…
Index Cond: (actual_departure = '2024-02-09'::date)
Index Searches: 1
Buffers: shared read=2
12 rows
Databases_kbtu > lab7 > lab7.sql
37:31 CRLF UTF-8 4 spaces

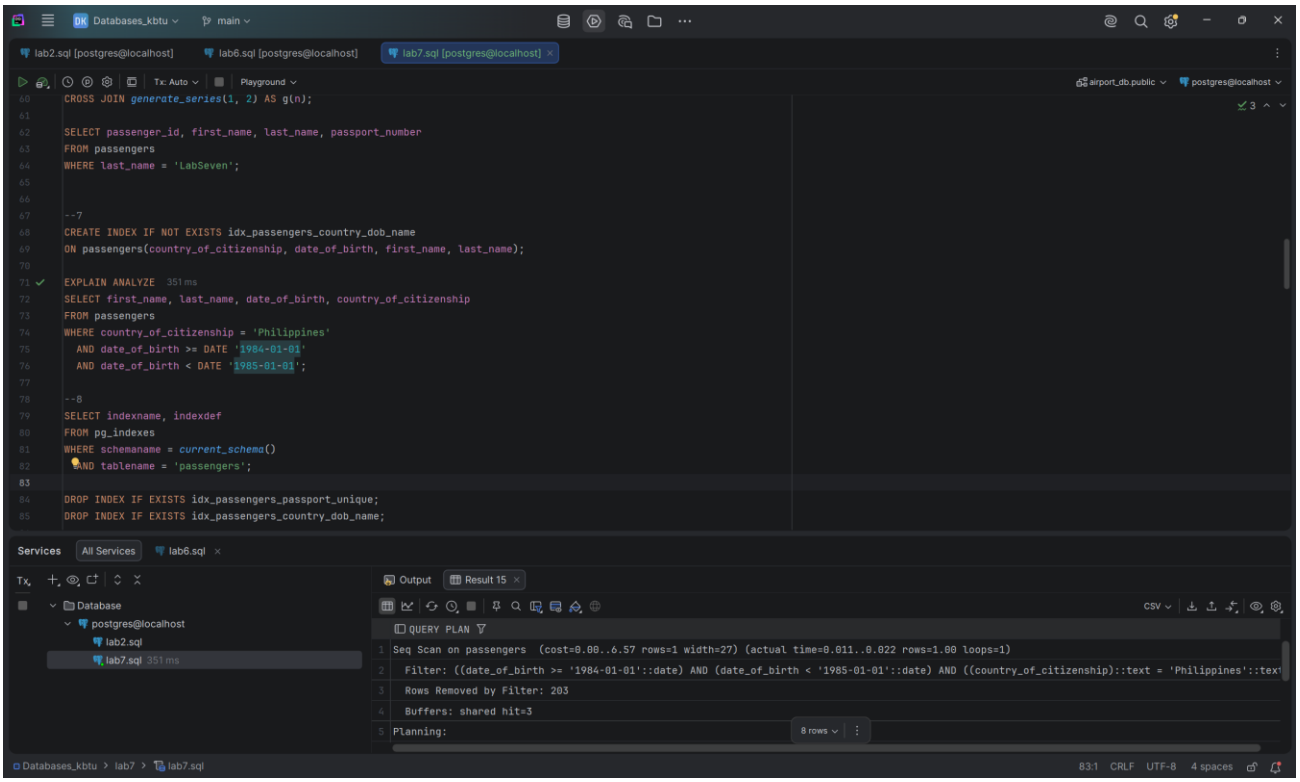
5—


FROM flights
WHERE actual_departure = DATE '2024-02-09';
--5
EXPLAIN ANALYZE
SELECT *
FROM flights
WHERE departure_airport_id = 1
AND arrival_airport_id = 2;
--6
CREATE UNIQUE INDEX IF NOT EXISTS idx_passengers_passport_unique
ON passengers(passport_number);
SELECT indexname, indexdef
FROM pg_indexes
WHERE schemaname = current_schema()
AND tablename = 'passengers'
AND indexname = 'idx_passengers_passport_unique';
INSERT INTO passengers
(passenger_id, first_name, last_name, date_of_birth,
gender, country_of_citizenship, country_of_residence, passport_number)
QUERY PLAN
Bitmap Heap Scan on flights (cost=4.30..9.97 rows=2 width=61) (actual time=0.057..0.058 rows=2.00
Recheck Cond: ((departure_airport_id = 1) AND (arrival_airport_id = 2))
Heap Blocks: exact=2
Buffers: shared hit=2 read=2
-> Bitmap Index Scan on idx_flights_dep_arr (cost=0.00..4.29 rows=2 width=0) (actual time=0.051
Index Cond: ((departure_airport_id = 1) AND (arrival_airport_id = 2))
Index Searches: 1
Buffers: shared read=2

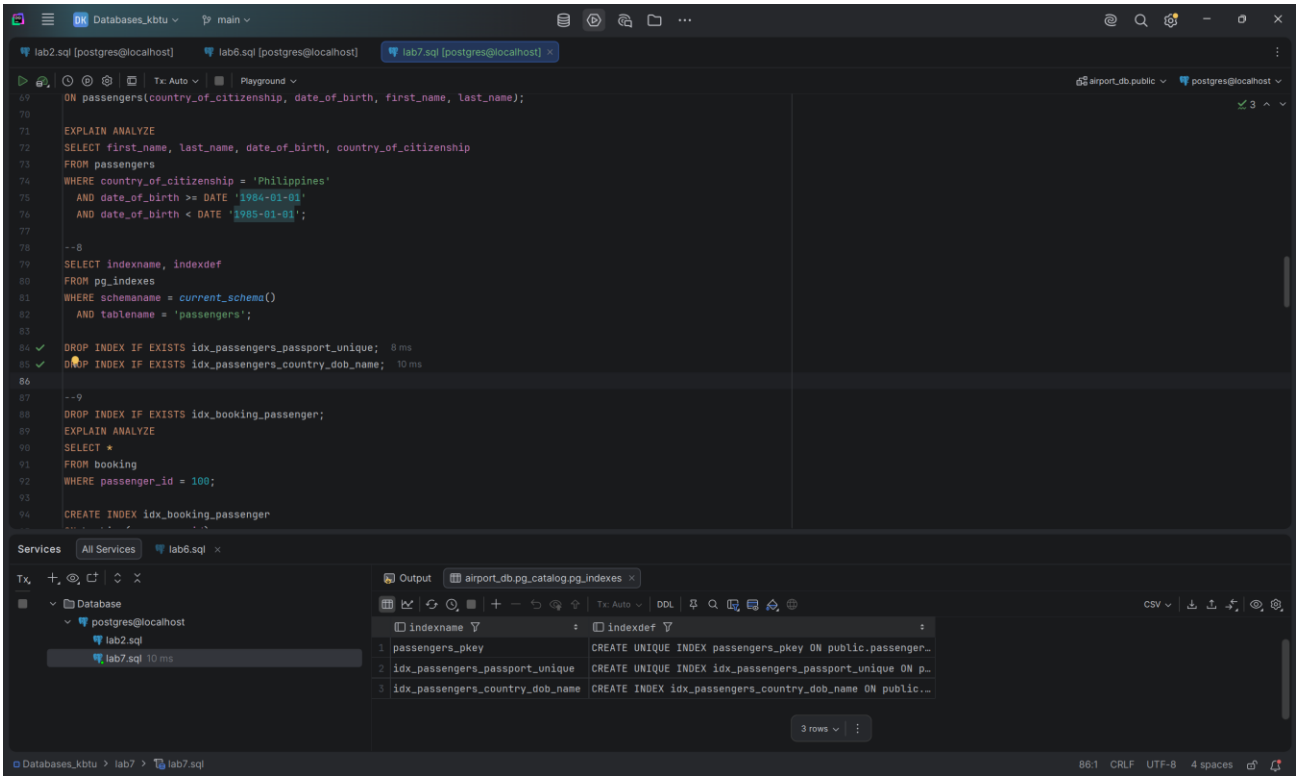
6—


--6
CREATE UNIQUE INDEX IF NOT EXISTS idx_passengers_passport_unique
ON passengers(passport_number);
SELECT indexname, indexdef
FROM pg_indexes
WHERE schemaname = current_schema()
AND tablename = 'passengers'
AND indexname = 'idx_passengers_passport_unique';
INSERT INTO passengers
(passenger_id, first_name, last_name, date_of_birth,
gender, country_of_citizenship, country_of_residence, passport_number)
SELECT passenger_id m.max_id + g.n,
CASE WHEN g.n = 1 THEN 'Almas' ELSE 'Aruzhan' END,
last_name 'LabSeven', DATE '2000-01-01',
CASE WHEN g.n = 1 THEN 'Male' ELSE 'Female' END,
country_of_citizenship 'Kazakhstan', country_of_residence 'Kazakhstan',
passport_number 'LAB7-' || (m.max_id + g.n)::text
FROM (SELECT COALESCE(MAX(passenger_id), 0) AS max_id FROM passengers) AS m
CROSS JOIN generate_series(1, 2) AS g(n);
SELECT passenger_id, first_name, last_name, passport_number 348 ms
FROM passengers
WHERE last_name = 'LabSeven';
passenger_id first_name last_name passport_number
201 Almas LabSeven LAB7-201
202 Aruzhan LabSeven LAB7-202
203 Almas LabSeven LAB7-203
204 Aruzhan LabSeven LAB7-204

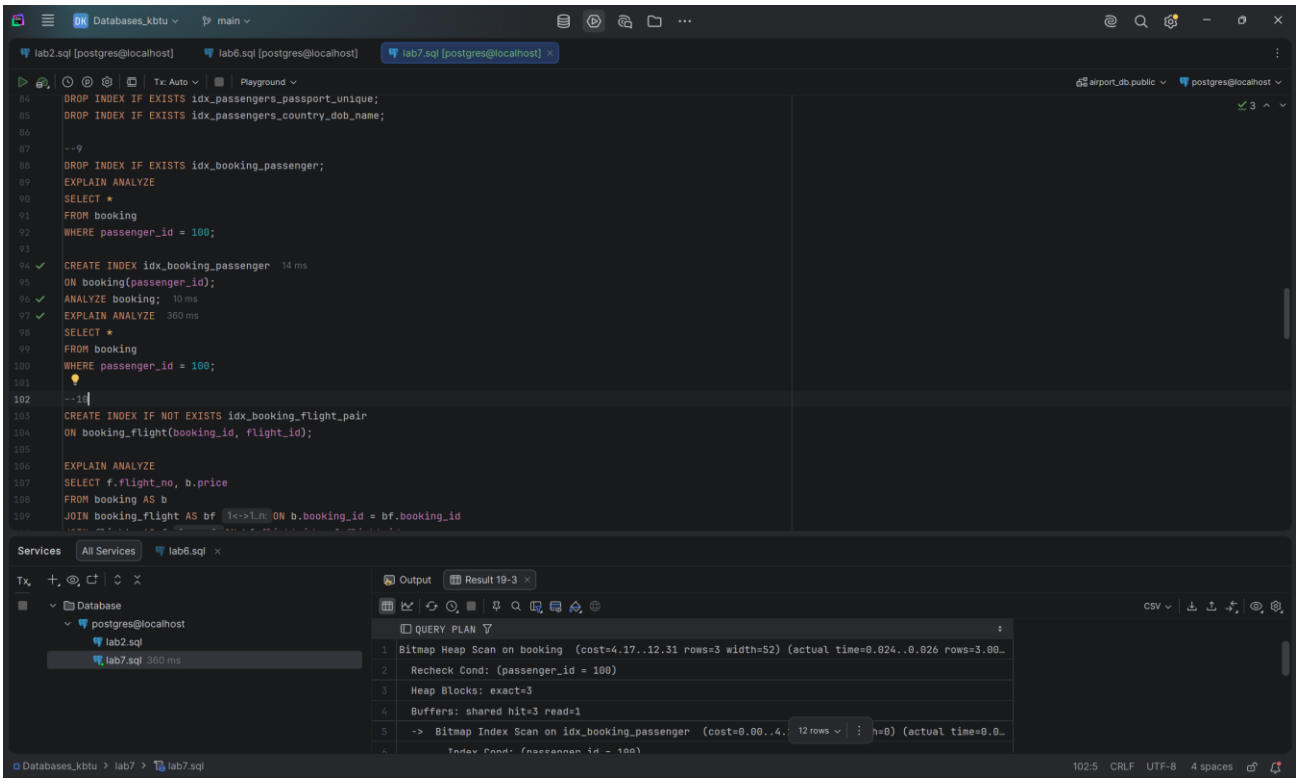
7—


CROSS JOIN generate_series(1, 2) AS g(n);
SELECT passenger_id, first_name, last_name, passport_number
FROM passengers
WHERE last_name = 'LabSeven';
--7
CREATE INDEX IF NOT EXISTS idx_passengers_country_dob_name
ON passengers(country_of_citizenship, date_of_birth, first_name, last_name);
EXPLAIN ANALYZE 351 ms
SELECT first_name, last_name, date_of_birth, country_of_citizenship
FROM passengers
WHERE country_of_citizenship = 'Philippines'
AND date_of_birth >= DATE '1984-01-01'
AND date_of_birth < DATE '1985-01-01';
--8
SELECT indexname, indexdef
FROM pg_indexes
WHERE schemaname = current_schema()
AND tablename = 'passengers';
DROP INDEX IF EXISTS idx_passengers_passport_unique;
DROP INDEX IF EXISTS idx_passengers_country_dob_name;
QUERY PLAN
Seq Scan on passengers (cost=0.00..6.57 rows=1 width=27) (actual time=0.011..0.022 rows=1.00 loops=1)
Filter: ((date_of_birth >= '1984-01-01'::date) AND (date_of_birth < '1985-01-01'::date) AND ((country_of_citizenship)::text = 'Philippines'::text
Rows Removed by Filter: 203
Buffers: shared hit=3
Planning:

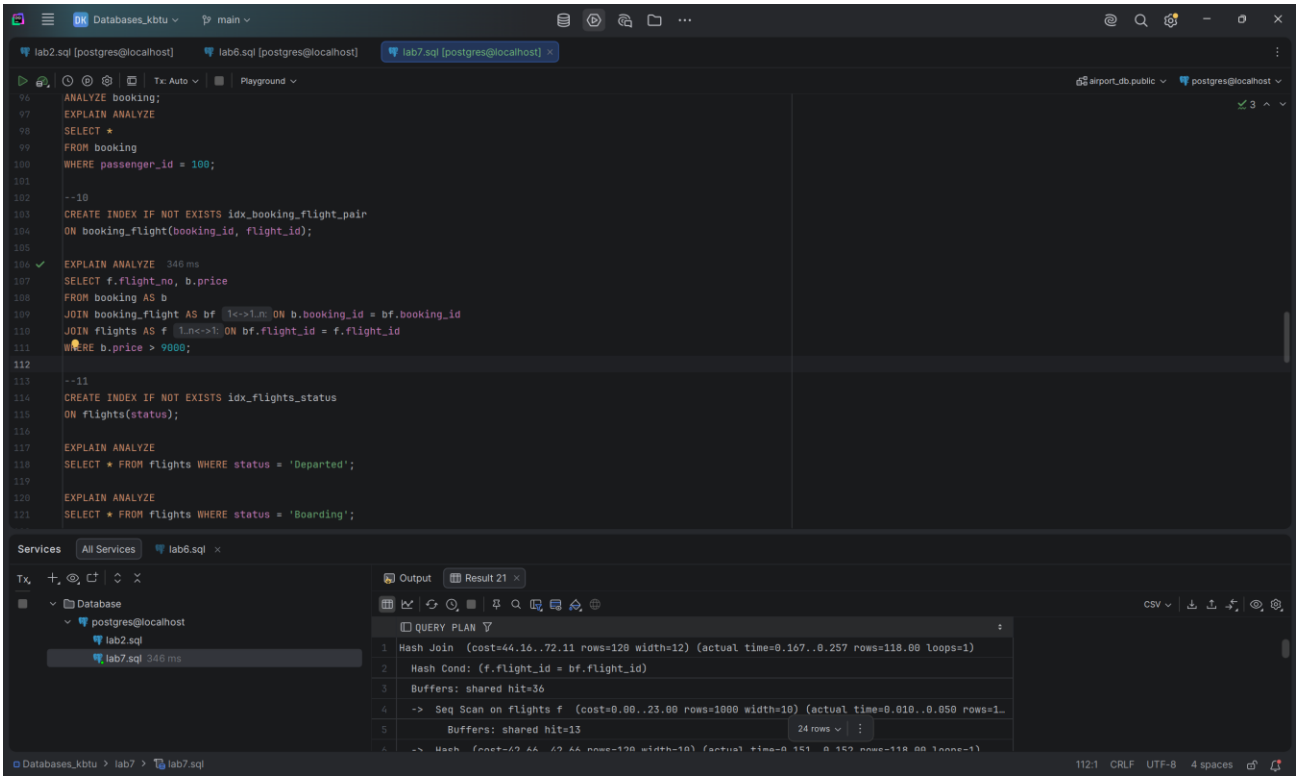
8—


ON passengers(country_of_citizenship, date_of_birth, first_name, last_name);
EXPLAIN ANALYZE
SELECT first_name, last_name, date_of_birth, country_of_citizenship
FROM passengers
WHERE country_of_citizenship = 'Philippines'
AND date_of_birth >= DATE '1984-01-01'
AND date_of_birth < DATE '1985-01-01';
--8
SELECT indexname, indexdef
FROM pg_indexes
WHERE schemaname = current_schema()
AND tablename = 'passengers';
DROP INDEX IF EXISTS idx_passengers_passport_unique; 8 ms
DROP INDEX IF EXISTS idx_passengers_country_dob_name; 10 ms
--9
DROP INDEX IF EXISTS idx_booking_passenger;
EXPLAIN ANALYZE
SELECT *
FROM booking
WHERE passenger_id = 100;
CREATE INDEX idx_booking_passenger
indexname
indexdef
passengers_pkey
CREATE UNIQUE INDEX passengers_pkey ON public.passenger…
idx_passengers_passport_unique
CREATE UNIQUE INDEX idx_passengers_passport_unique ON p…
idx_passengers_country_dob_name
CREATE INDEX idx_passengers_country_dob_name ON public.…
3 rows

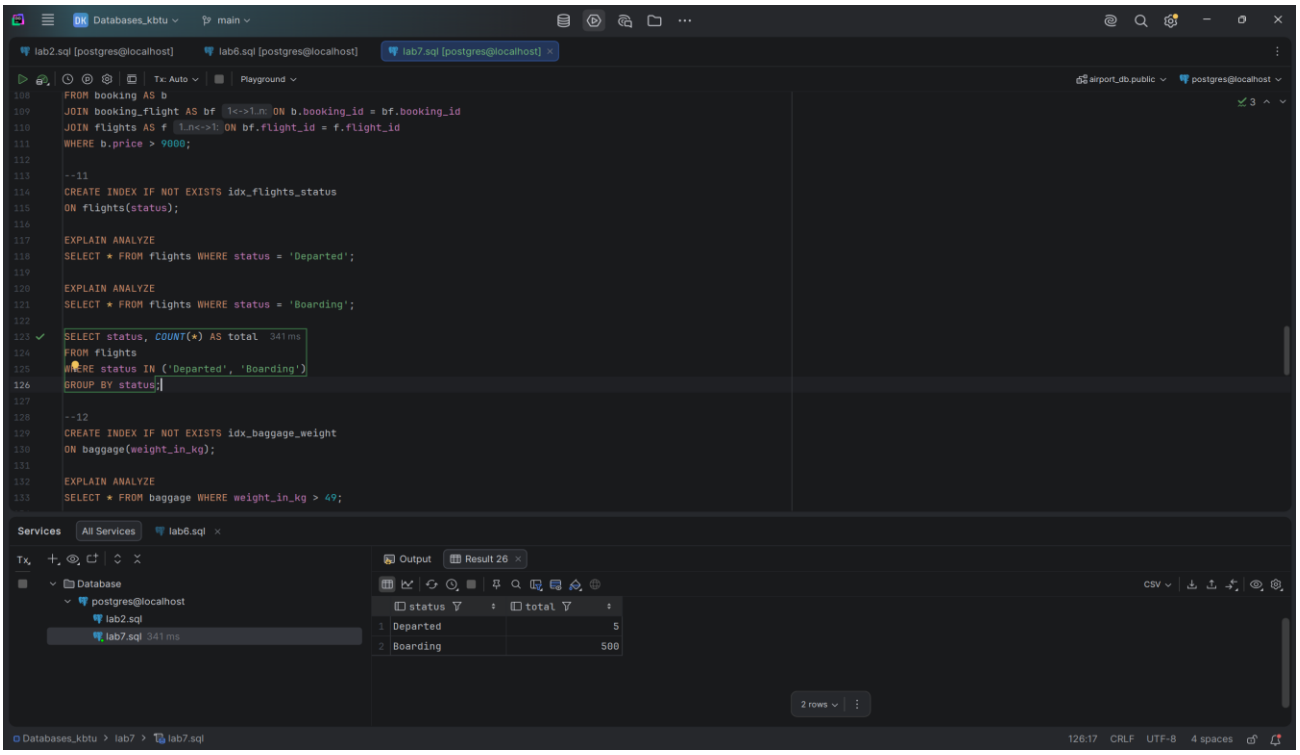
9—


DROP INDEX IF EXISTS idx_passengers_passport_unique;
DROP INDEX IF EXISTS idx_passengers_country_dob_name;
--9
DROP INDEX IF EXISTS idx_booking_passenger;
EXPLAIN ANALYZE
SELECT *
FROM booking
WHERE passenger_id = 100;
CREATE INDEX idx_booking_passenger 14 ms
ON booking(passenger_id);
ANALYZE booking; 10 ms
EXPLAIN ANALYZE 360 ms
SELECT *
FROM booking
WHERE passenger_id = 100;
--10
CREATE INDEX IF NOT EXISTS idx_booking_flight_pair
ON booking_flight(booking_id, flight_id);
EXPLAIN ANALYZE
SELECT f.flight_no, b.price
FROM booking AS b
JOIN booking_flight AS bf
ON b.booking_id = bf.booking_id
QUERY PLAN
Bitmap Heap Scan on booking (cost=4.17..12.31 rows=3 width=52) (actual time=0.024..0.026 rows=3.00
Recheck Cond: (passenger_id = 100)
Heap Blocks: exact=3
Buffers: shared hit=3 read=1
-> Bitmap Index Scan on idx_booking_passenger (cost=0.00..4.
12 rows

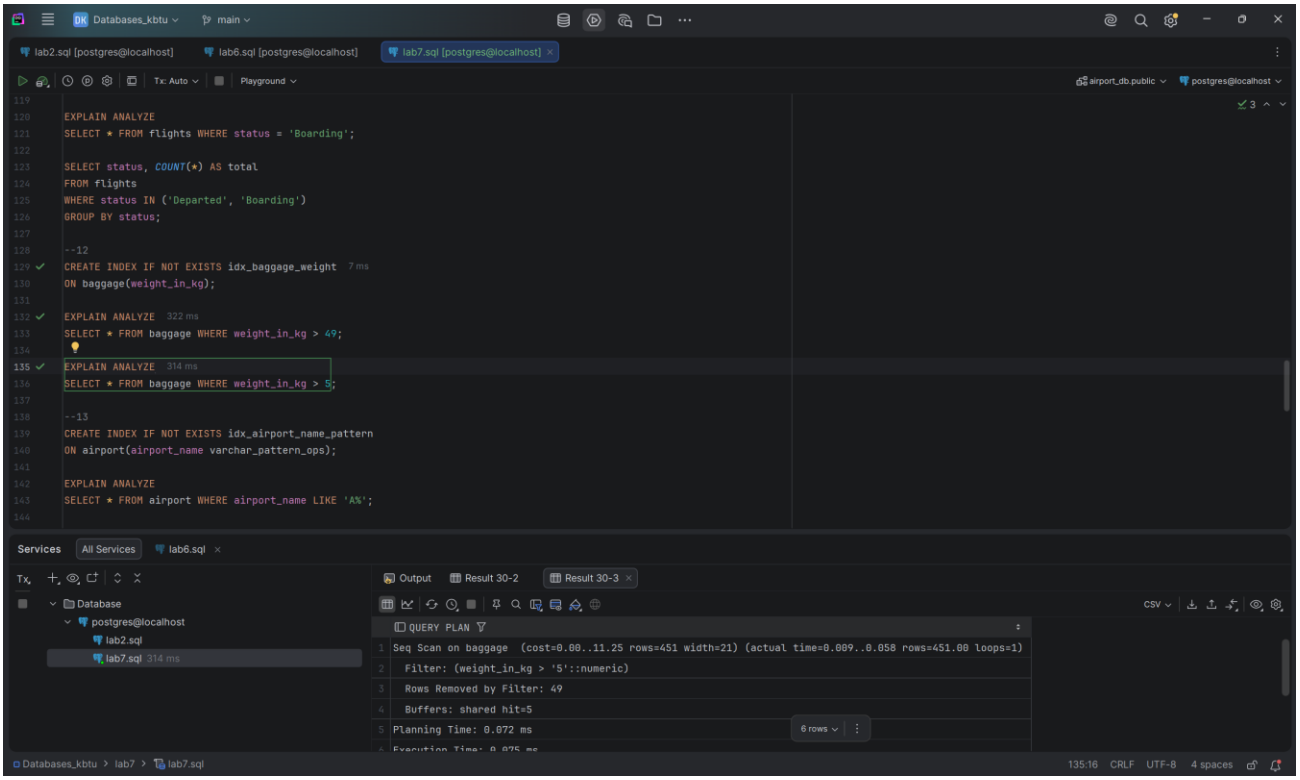
--10


ANALYZE booking;
EXPLAIN ANALYZE
SELECT *
FROM booking
WHERE passenger_id = 100;
--10
CREATE INDEX IF NOT EXISTS idx_booking_flight_pair
ON booking_flight(booking_id, flight_id);
EXPLAIN ANALYZE 346 ms
SELECT f.flight_no, b.price
FROM booking AS b
JOIN booking_flight AS bf
ON b.booking_id = bf.booking_id
JOIN flights AS f
ON bf.flight_id = f.flight_id
WHERE b.price > 9000;
--11
CREATE INDEX IF NOT EXISTS idx_flights_status
ON flights(status);
EXPLAIN ANALYZE
SELECT * FROM flights WHERE status = 'Departed';
EXPLAIN ANALYZE
SELECT * FROM flights WHERE status = 'Boarding';
QUERY PLAN
Hash Join (cost=44.16..72.11 rows=120 width=12) (actual time=0.167..0.257 rows=118.00 loops=1)
Hash Cond: (f.flight_id = bf.flight_id)
Buffers: shared hit=36
-> Seq Scan on flights f (cost=0.00..23.00 rows=1000 width=10) (actual time=0.010..0.050 rows=1
Buffers: shared hit=13
24 rows

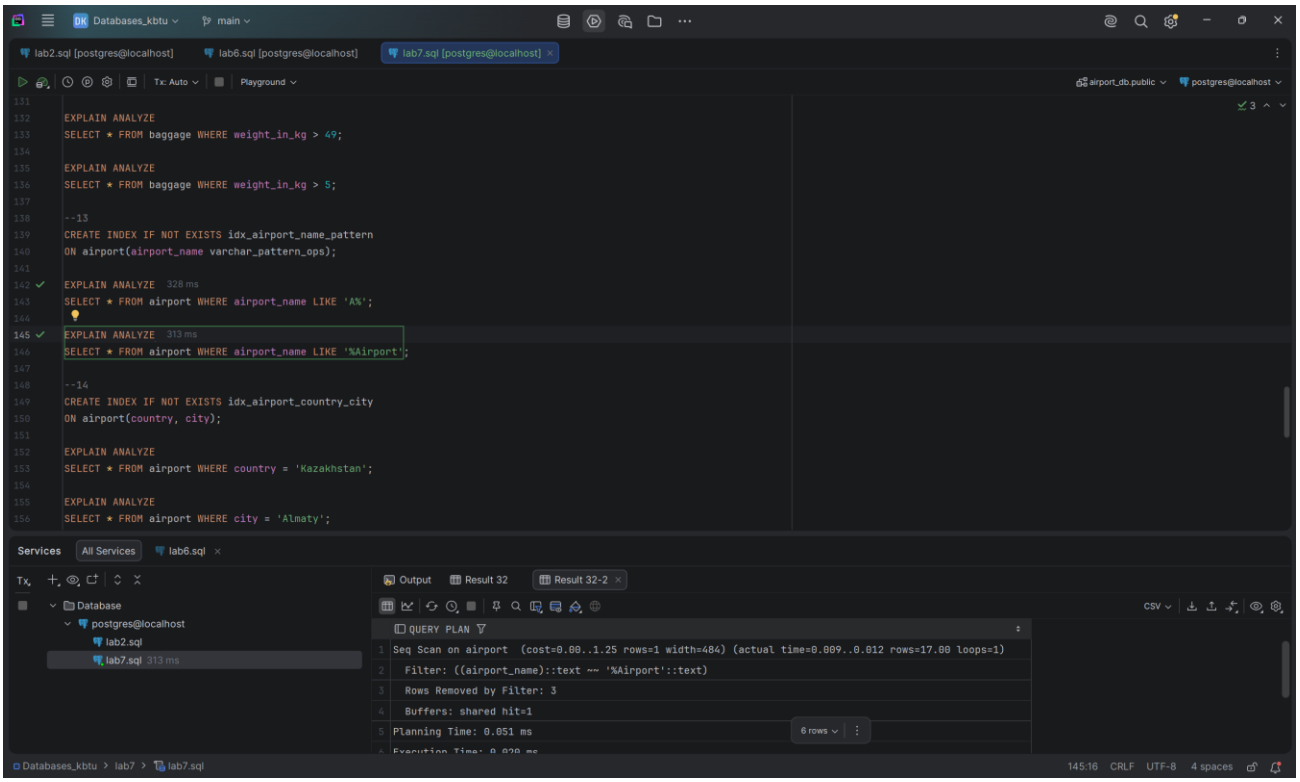
--11


Databases_kbtu
main
lab2.sql [postgres@localhost]
lab6.sql [postgres@localhost]
lab7.sql [postgres@localhost]
FROM booking AS b
JOIN booking_flight AS bf ON b.booking_id = bf.booking_id
JOIN flights AS f ON bf.flight_id = f.flight_id
WHERE b.price > 9000;
--11
CREATE INDEX IF NOT EXISTS idx_flights_status
ON flights(status);
EXPLAIN ANALYZE
SELECT * FROM flights WHERE status = 'Departed';
EXPLAIN ANALYZE
SELECT * FROM flights WHERE status = 'Boarding';
SELECT status, COUNT(*) AS total 341 ms
FROM flights
WHERE status IN ('Departed', 'Boarding')
GROUP BY status;
--12
CREATE INDEX IF NOT EXISTS idx_baggage_weight
ON baggage(weight_in_kg);
EXPLAIN ANALYZE
SELECT * FROM baggage WHERE weight_in_kg > 49;
Services
All Services
lab6.sql
Output
Result 26
status
total
Departed 5
Boarding 500
2 rows
Database
postgres@localhost
lab2.sql
lab7.sql 341 ms
126:17 CRLF UTF-8 4 spaces

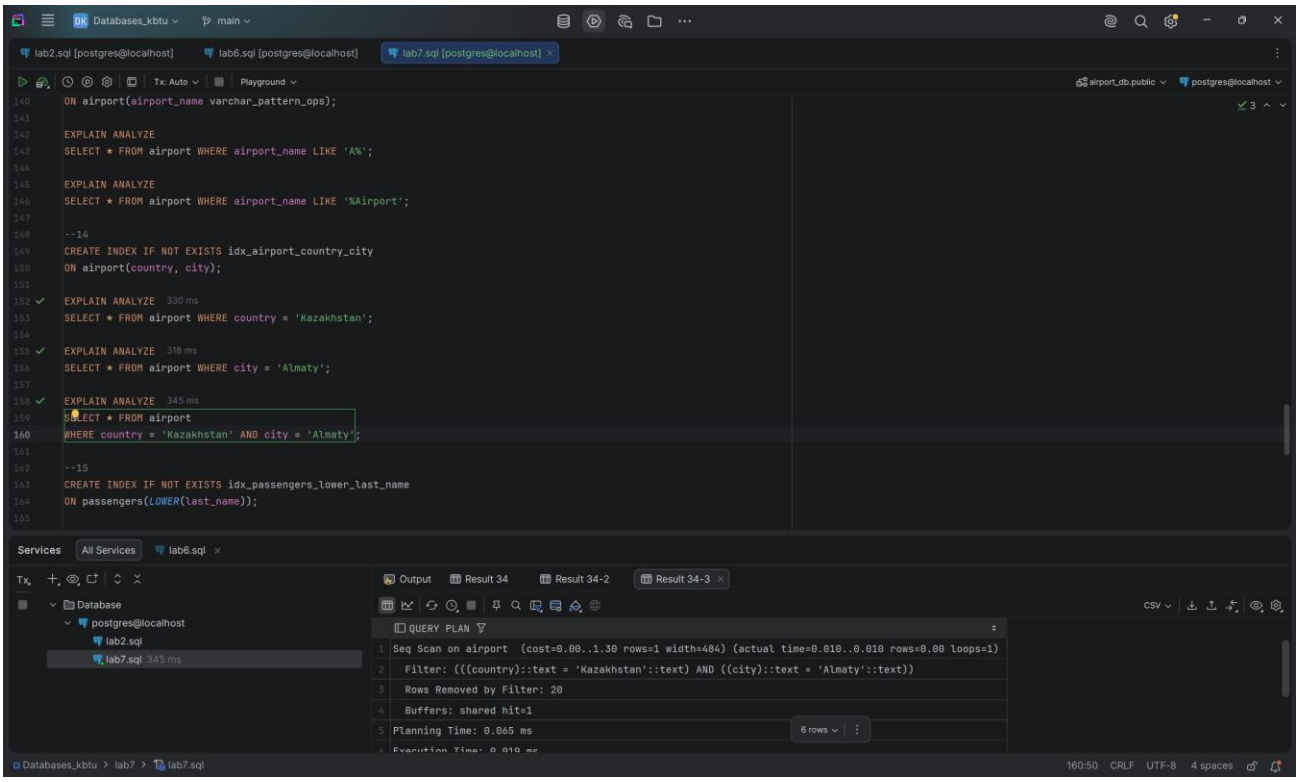
--12


Databases_kbtu
main
lab2.sql [postgres@localhost]
lab6.sql [postgres@localhost]
lab7.sql [postgres@localhost]
EXPLAIN ANALYZE
SELECT * FROM flights WHERE status = 'Boarding';
SELECT status, COUNT(*) AS total
FROM flights
WHERE status IN ('Departed', 'Boarding')
GROUP BY status;
--12
CREATE INDEX IF NOT EXISTS idx_baggage_weight 7 ms
ON baggage(weight_in_kg);
EXPLAIN ANALYZE 322 ms
SELECT * FROM baggage WHERE weight_in_kg > 49;
EXPLAIN ANALYZE 314 ms
SELECT * FROM baggage WHERE weight_in_kg > 5;
--13
CREATE INDEX IF NOT EXISTS idx_airport_name_pattern
ON airport(airport_name varchar_pattern_ops);
EXPLAIN ANALYZE
SELECT * FROM airport WHERE airport_name LIKE 'A%';
Services
All Services
lab6.sql
Output
Result 30-2
Result 30-3
QUERY PLAN
Seq Scan on baggage (cost=0.00..11.25 rows=451 width=21) (actual time=0.089..0.058 rows=451.00 loops=1)
Filter: (weight_in_kg > '5'::numeric)
Rows Removed by Filter: 49
Buffers: shared hit=5
Planning Time: 0.072 ms
6 rows
Database
postgres@localhost
lab2.sql
lab7.sql 314 ms
135:16 CRLF UTF-8 4 spaces

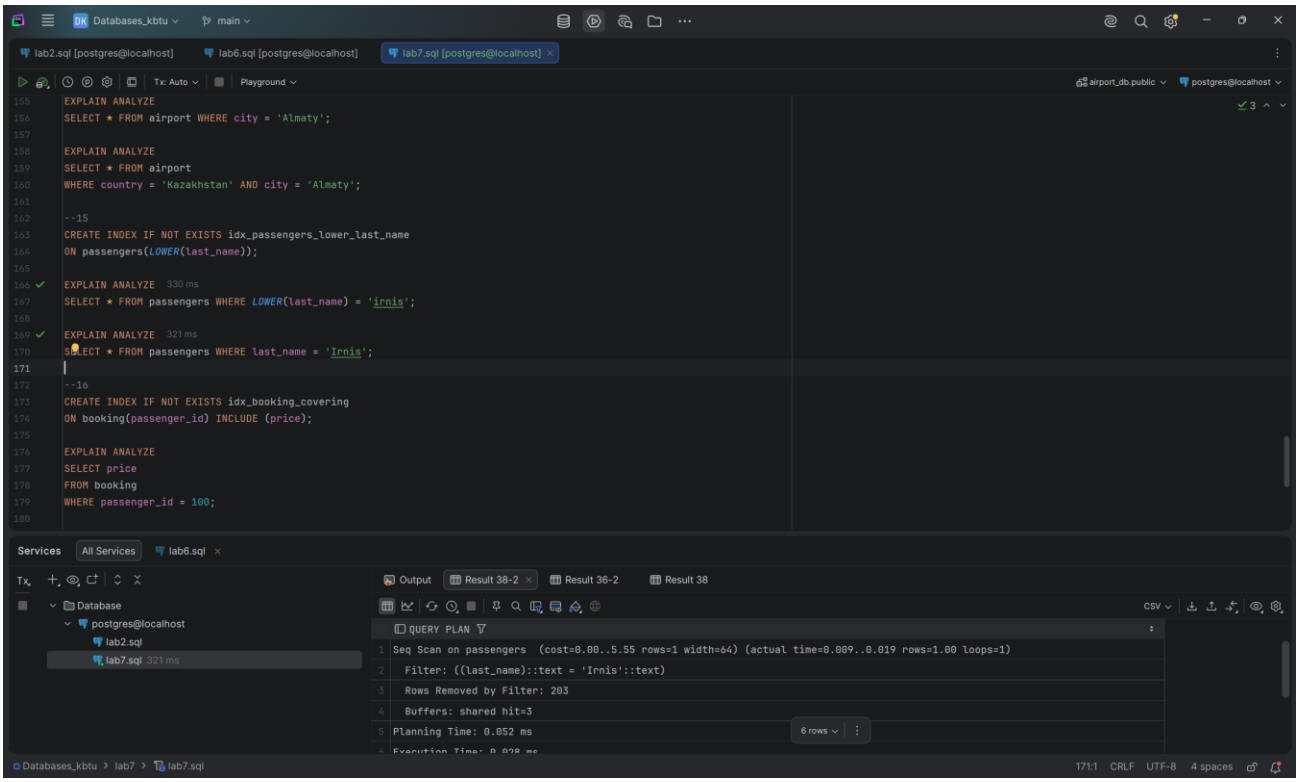
--13


EXPLAIN ANALYZE
SELECT * FROM baggage WHERE weight_in_kg > 49;
EXPLAIN ANALYZE
SELECT * FROM baggage WHERE weight_in_kg > 5;
--13
CREATE INDEX IF NOT EXISTS idx_airport_name_pattern
ON airport(airport_name varchar_pattern_ops);
EXPLAIN ANALYZE 328 ms
SELECT * FROM airport WHERE airport_name LIKE 'A%';
EXPLAIN ANALYZE 313 ms
SELECT * FROM airport WHERE airport_name LIKE '%Airport';
--14
CREATE INDEX IF NOT EXISTS idx_airport_country_city
ON airport(country, city);
EXPLAIN ANALYZE
SELECT * FROM airport WHERE country = 'Kazakhstan';
EXPLAIN ANALYZE
SELECT * FROM airport WHERE city = 'Almaty';
QUERY PLAN
Seq Scan on airport (cost=0.00..1.25 rows=1 width=484) (actual time=0.009..0.012 rows=17.00 loops=1)
Filter: ((airport_name)::text ~~ '%Airport'::text)
Rows Removed by Filter: 3
Buffers: shared hit=1
Planning Time: 0.051 ms

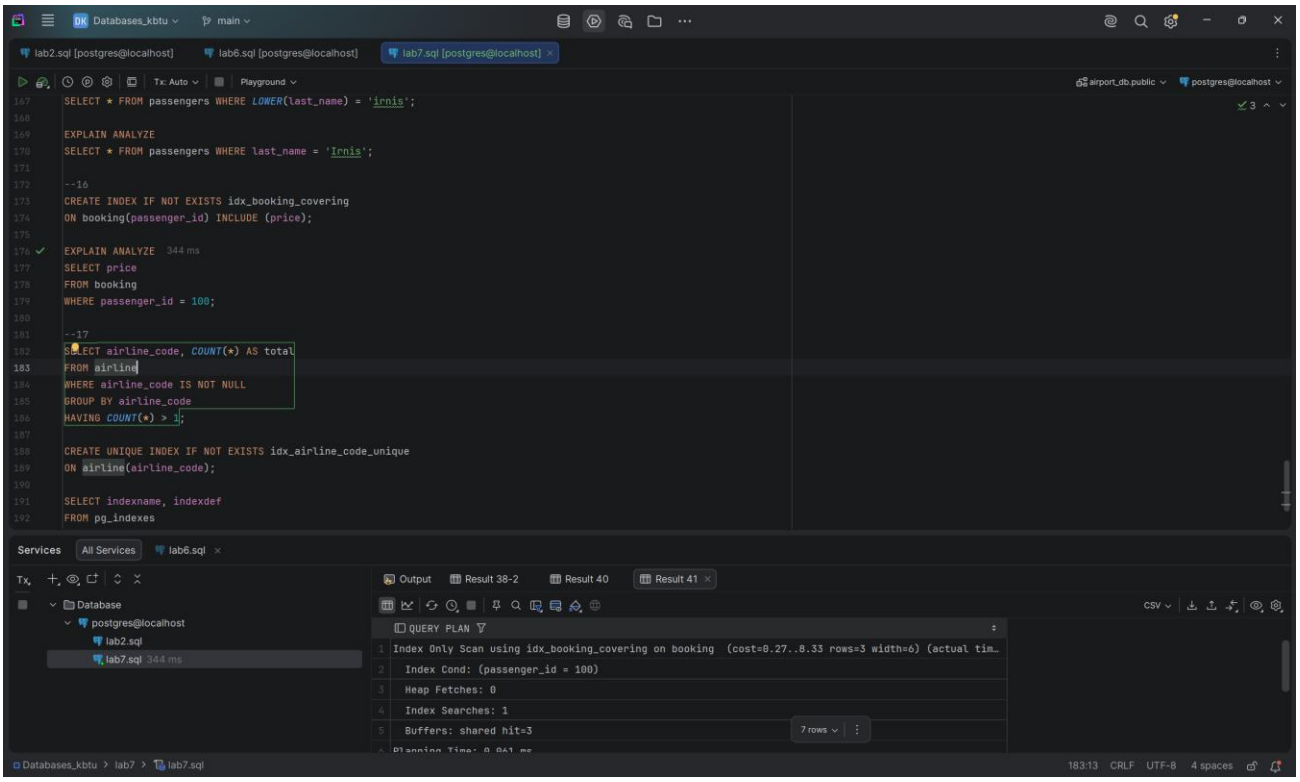
--14


ON airport(airport_name varchar_pattern_ops);
EXPLAIN ANALYZE
SELECT * FROM airport WHERE airport_name LIKE 'A%';
EXPLAIN ANALYZE
SELECT * FROM airport WHERE airport_name LIKE '%Airport';
--14
CREATE INDEX IF NOT EXISTS idx_airport_country_city
ON airport(country, city);
EXPLAIN ANALYZE 330 ms
SELECT * FROM airport WHERE country = 'Kazakhstan';
EXPLAIN ANALYZE 318 ms
SELECT * FROM airport WHERE city = 'Almaty';
EXPLAIN ANALYZE 345 ms
SELECT * FROM airport
WHERE country = 'Kazakhstan' AND city = 'Almaty';
--15
CREATE INDEX IF NOT EXISTS idx_passengers_lower_last_name
ON passengers(LOWER(last_name));
QUERY PLAN
Seq Scan on airport (cost=0.00..1.30 rows=1 width=484) (actual time=0.010..0.010 rows=0.00 loops=1)
Filter: (((country)::text = 'Kazakhstan'::text) AND ((city)::text = 'Almaty'::text))
Rows Removed by Filter: 20
Buffers: shared hit=1
Planning Time: 0.065 ms

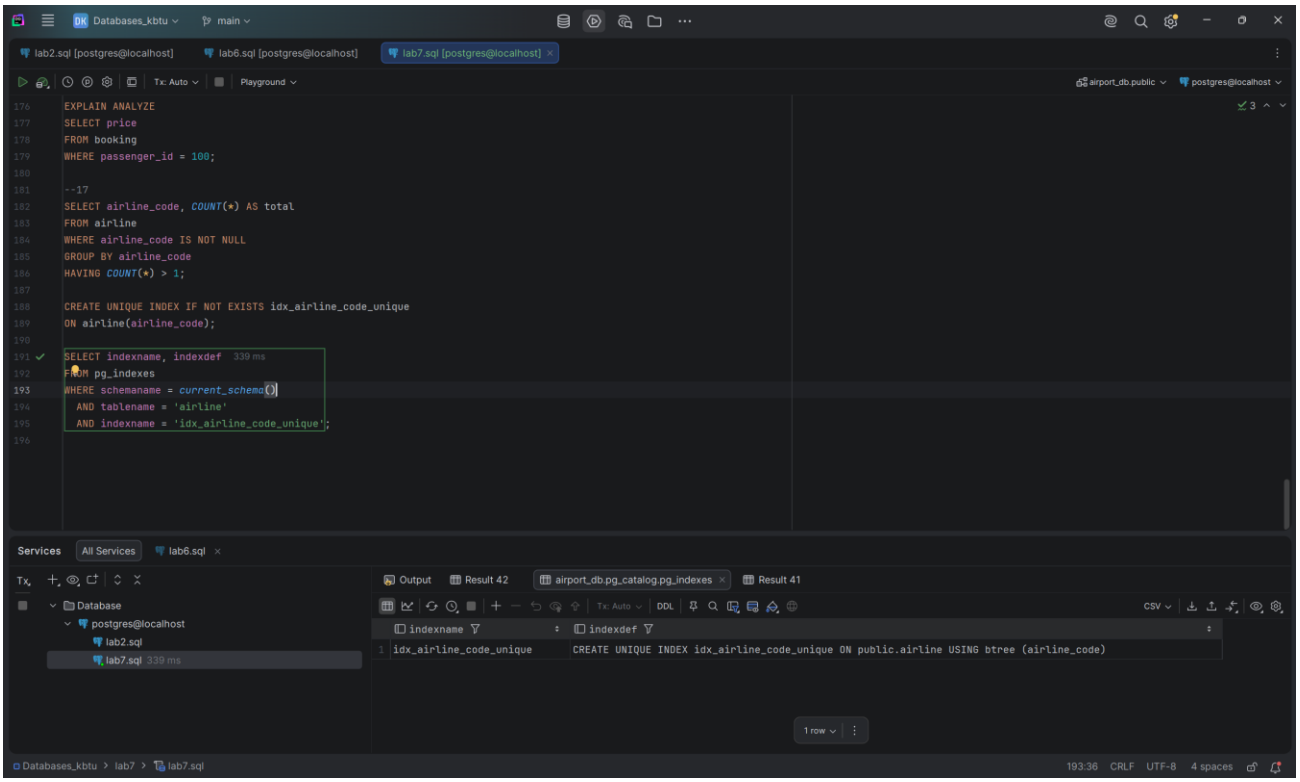
--15


Databases_kbtu
main
lab2.sql [postgres@localhost]
lab6.sql [postgres@localhost]
lab7.sql [postgres@localhost]
EXPLAIN ANALYZE
SELECT * FROM airport WHERE city = 'Almaty';
EXPLAIN ANALYZE
SELECT * FROM airport
WHERE country = 'Kazakhstan' AND city = 'Almaty';
--15
CREATE INDEX IF NOT EXISTS idx_passengers_lower_last_name
ON passengers(LOWER(last_name));
EXPLAIN ANALYZE 330 ms
SELECT * FROM passengers WHERE LOWER(last_name) = 'irnis';
EXPLAIN ANALYZE 321 ms
SELECT * FROM passengers WHERE last_name = 'Irnis';
--16
CREATE INDEX IF NOT EXISTS idx_booking_covering
ON booking(passenger_id) INCLUDE (price);
EXPLAIN ANALYZE
SELECT price
FROM booking
WHERE passenger_id = 100;
Services
All Services
lab6.sql
Output
Result 38-2
Result 36-2
Result 38
QUERY PLAN
Seq Scan on passengers (cost=0.00..5.55 rows=1 width=64) (actual time=0.009..0.019 rows=1.00 loops=1)
Filter: ((last_name)::text = 'Irnis'::text)
Rows Removed by Filter: 203
Buffers: shared hit=3
Planning Time: 0.052 ms
6 rows
Databases_kbtu > lab7 > lab7.sql
171:1 CRLF UTF-8 4 spaces

--16


Databases_kbtu
main
lab2.sql [postgres@localhost]
lab6.sql [postgres@localhost]
lab7.sql [postgres@localhost]
SELECT * FROM passengers WHERE LOWER(last_name) = 'irnis';
EXPLAIN ANALYZE
SELECT * FROM passengers WHERE last_name = 'Irnis';
--16
CREATE INDEX IF NOT EXISTS idx_booking_covering
ON booking(passenger_id) INCLUDE (price);
EXPLAIN ANALYZE 344 ms
SELECT price
FROM booking
WHERE passenger_id = 100;
--17
SELECT airline_code, COUNT(*) AS total
FROM airline
WHERE airline_code IS NOT NULL
GROUP BY airline_code
HAVING COUNT(*) > 1;
CREATE UNIQUE INDEX IF NOT EXISTS idx_airline_code_unique
ON airline(airline_code);
SELECT indexname, indexdef
FROM pg_indexes
Services
All Services
lab6.sql
Output
Result 38-2
Result 40
Result 41
QUERY PLAN
Index Only Scan using idx_booking_covering on booking (cost=0.27..8.33 rows=3 width=6) (actual tim…
Index Cond: (passenger_id = 100)
Heap Fetches: 0
Index Searches: 1
Buffers: shared hit=3
7 rows
Databases_kbtu > lab7 > lab7.sql
183:13 CRLF UTF-8 4 spaces

--17


Databases_kbtu
main
lab2.sql [postgres@localhost]
lab6.sql [postgres@localhost]
lab7.sql [postgres@localhost]
Tx: Auto
Playground
airport_db.public
postgres@localhost
EXPLAIN ANALYZE
SELECT price
FROM booking
WHERE passenger_id = 100;
--17
SELECT airline_code, COUNT(*) AS total
FROM airline
WHERE airline_code IS NOT NULL
GROUP BY airline_code
HAVING COUNT(*) > 1;
CREATE UNIQUE INDEX IF NOT EXISTS idx_airline_code_unique
ON airline(airline_code);
SELECT indexname, indexdef 339 ms
FROM pg_indexes
WHERE schemaname = current_schema()
AND tablename = 'airline'
AND indexname = 'idx_airline_code_unique';
Services
All Services
lab6.sql
Database
postgres@localhost
lab2.sql
lab7.sql 339 ms
Output
Result 42
airport_db.pg_catalog.pg_indexes
Result 41
DDL
CSV
indexname
indexdef
idx_airline_code_unique
CREATE UNIQUE INDEX idx_airline_code_unique ON public.airline USING btree (airline_code)
1 row
Databases_kbtu > lab7 > lab7.sql
193:36 CRLF UTF-8 4 spaces